

ALEKSEI LIAN

Toronto, ON

(343) 987-7827 | alekseilianv@gmail.com | linkedin.com/in/aleksei-lian | github.com/alxlyn | alekseilian.com

SUMMARY

Computer Science student at York University with strong foundations in systems programming, algorithms, and backend development. Experience building production-grade Python services with FastAPI, PostgreSQL, Redis, and Docker, deployed via CI/CD on Google Cloud. Co-Founder and lead technical contributor at an early-stage startup, identifying and resolving concurrency race conditions in production middleware.

EDUCATION

York University

Jan 2025 – Present

Bachelor of Science in Computer Science (Expected Spring 2027)

Toronto, ON

Carleton University (Transferred)

Sep 2023 – Dec 2024

Bachelor of Science in Computer Science

Ottawa, ON

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Discrete Math, Operating Systems

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, JavaScript

Frameworks & Databases: Flask, FastAPI, REST APIs, PostgreSQL, SQLAlchemy

Infrastructure & Tools: Docker, GCP (Cloud Run, Cloud Build), Linux, Git, CI/CD, Cursor, Claude Code, GitHub Copilot

Concepts: Data Structures & Algorithms, Concurrency, OOP, Distributed Systems

PROJECTS

URL Shortener | Python, FastAPI, asyncpg, PostgreSQL, Redis, Docker

2026

- Engineered an async URL shortener with **FastAPI** and **asyncpg**, leveraging connection pooling to handle concurrent I/O; validated against 100,000 unique short codes with zero collisions.
- Built **RESTful APIs** for link creation, redirection, click-tracking, and a leaderboard endpoint; enforced per-IP **rate limiting** via **slowapi** + **Redis** (GCP load-balancer-aware).
- Automated CI with **GitHub Actions** (lint + 34 **pytest** integration tests against live Postgres) and CD via **Google Cloud Build**, deploying to Cloud Run with zero-downtime rollouts.
- Generated cryptographically secure short codes via Python's `secrets` module with server-side URL validation and DB-level uniqueness constraints; retry logic prevents collisions at scale.

Chess Engine (UCI Compatible) | Python

2026

- Implemented **Negamax** with **alpha-beta pruning** and **iterative deepening**, reaching depth 5–6 in 0.5–1.5 s midgame; integrated with **Banksia** and **Arena** GUIs via the UCI protocol.
- Implemented Quiescence search to resolve tactical "horizon effects" by evaluating captures and checks before returning static scores.
- Added a Transposition Table with FEN-based hashing to cache evaluated positions, cutting redundant computation across the search tree.
- Applied **MVV-LVA** move ordering to prioritize high-impact captures and deadline-based time management to stay within UCI time controls across all game phases.

EXPERIENCE

Tanish (Early-Stage Startup)

January 2026 – Present

Co-Founder & Lead Backend Engineer

Bishkek, Kyrgyzstan

- Built a full-stack dating platform using FastAPI, PostgreSQL, and SQLAlchemy with JWT authentication, real-time WebSocket chat, mutual-match detection, and photo uploads.
- Identified and resolved concurrency race conditions in Python middleware that caused intermittent data failures under parallel load.
- Led technical delivery for 2 contracted engineers building the core mobile app; enforced code standards via pull request reviews, cutting post-merge bugs by reducing integration issues.

Independent Tutoring

2025 – Present

Mathematics Tutor

Toronto, ON

- Provide individualized instruction in mathematics, specializing in breaking down complex spatial concepts such as geometry and coordinate transformations.

KPD Advertising Firm**May 2022 – August 2025 (Seasonal)***Production and IT Support Assistant**Bishkek, Kyrgyzstan*

- Set up centralized file-sharing using SMB/NFS, reducing data transfer time by 50% across the office network.
- Maintained network and hardware infrastructure for 5+ workstations, troubleshooting issues to keep 99.9% uptime.
- Performed OS installations, software setup, and system optimization across office workstations; trained non-technical staff on the centralized file-sharing system.